

Deep Learning - Theory and Practice

IE 643
Lecture 9

August 28, 2025.

- 1 MLP for multi-class classification
 - Cross-entropy
 - Training MLP for multi-class classification

- 2 MLP for multi-label classification

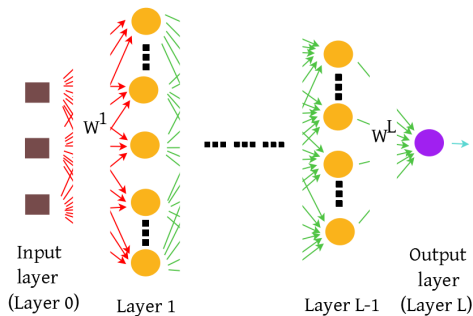
MLP for multi-class classification

- **Input:** Training Data $D = \{(x^i, y^i)\}_{i=1}^S$, $x^i \in \mathcal{X} \subseteq \mathbb{R}^d$, $y^i \in \mathcal{Y}$, $\forall i \in \{1, \dots, S\}$ and MLP architecture parametrized by weights w .
- **New Task:** $\mathcal{Y} = \{1, \dots, C\}$, $C \geq 2$.
- Corresponds to multi-class classification.

Question 1: What is a suitable architecture for the MLP's last (or output) layer?

Question 2: What is a suitable loss (or error) function?

MLP for multi-class classification



Question 1: Can the same MLP architecture with single output neuron used in binary classification be used for multi-class classification?

Question 2: Can the same logistic sigmoidal activation function for the output neuron used in binary classification be used for multi-class classification?

MLP for multi-class classification

We will use the following approach for multi-class classification:

- **Input:** Training Data $D = \{(x^i, y^i)\}_{i=1}^S$, $x^i \in \mathcal{X} \subseteq \mathbb{R}^d$, $y^i \in \mathcal{Y}$, $\forall i \in \{1, \dots, S\}$ and MLP architecture parametrized by weights w .
- **New Task:** $\mathcal{Y} = \{1, \dots, C\}$, $C \geq 2$ corresponds to multi-class classification.

- Transform $y = c$ to $y^{\text{onehotenc}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$.

- **Note:** $y^{\text{onehotenc}} \in \{0, 1\}^C$ corresponding to $y = c \in \mathcal{Y}$ has a 1 at c -th coordinate, and other entries as zeros.
- $y^{\text{onehotenc}}$ is called the **one-hot encoding** of y .

MLP for multi-class classification

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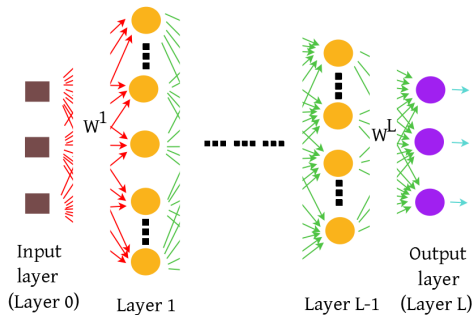
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- $y^{onehotenc}$ is called the **one-hot encoding** of y .
- $y^{onehotenc}$ for $y = c$ corresponds to a **discrete probability distribution** with its entire mass concentrated at the c -th coordinate.

MLP for multi-class classification

What change can be made to the network architecture so that the MLP outputs a discrete probability distribution?

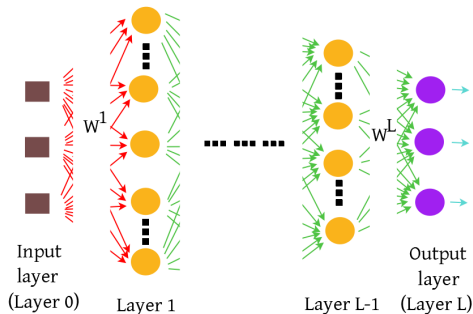
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MLP for multi-class classification

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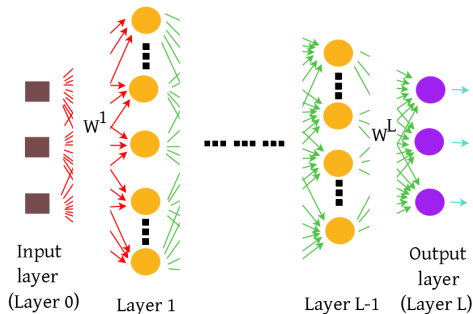


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MLP for multi-class classification

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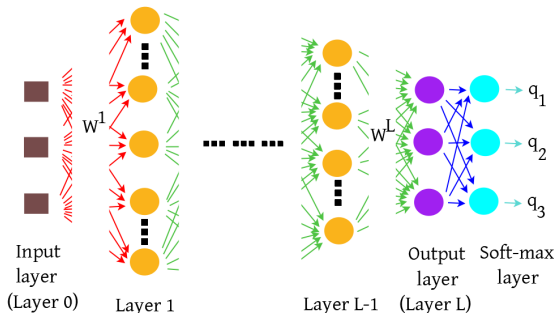
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MLP for multi-class classification

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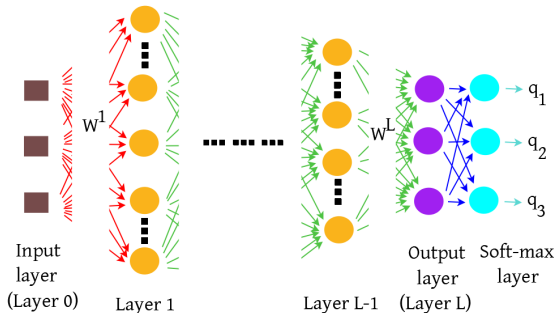
Step 2: Perform a soft-max function over the outputs from output layer so that the outputs are transformed into probabilities.



MLP for multi-class classification

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What is a soft-max function?

MLP for multi-class classification

What is a soft-max function?

- Given arbitrary activations $a_1^L, a_2^L, \dots, a_C^L$ from an output layer (L -th layer), how do we get probabilities?
- Perform the following transformation:

$$q_j = \frac{\exp(a_j^L)}{\sum_{r=1}^C \exp(a_r^L)}, \quad \forall j = 1, \dots, C.$$

- $q_1, \dots, q_C$ form a discrete probability distribution. **(Verify this claim!)**

The transformation used to obtain the probabilities q_j is called the soft-max function.

MLP for multi-class classification

Now that the MLP outputs a discrete probability distribution, how do we compare the one-hot encoding and the output distribution?

- We will use the popular divergence measure called **Kullback-Liebler** divergence (or KL-divergence).
- Given two discrete probability distributions $p = (p_1, \dots, p_C)$ and $q = (q_1, \dots, q_C)$, where $q_j > 0 \forall j = 1, \dots, C$, KL-divergence between p and q is defined as:

$$KL(p||q) = \sum_{j=1}^C p_j \log \frac{p_j}{q_j}.$$

- **Note:** The distribution p is usually called the **true** distribution and the distribution q is called the **predicted** distribution.
- Does the soft-max function give predictions $q_j > 0, j = 1, \dots, C$?

MLP for multi-class classification: KL Divergence

Some useful properties of KL-divergence:

- KL-divergence is **not** a distance function.

MLP for multi-class classification: KL Divergence

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- KL-divergence is **not** a distance function.
- **Recall:** A distance function $d : X \times X \rightarrow [0, \infty)$ has the following properties:
 - ▶ $d(x, x) = 0, \forall x \in X$ (identity of indistinguishables)
 - ▶ $d(x, y) = d(y, x), \forall x, y \in X$ (Symmetry)
 - ▶ $d(x, z) \leq d(x, y) + d(y, z), \forall x, y, z \in X$ (triangle inequality)

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- KL-divergence does not obey symmetry property.
 - ▶ Simple example: compute $KL(p||q)$ and $KL(q||p)$ for $p = (1/4, 3/4)$ and $q = (1/2, 1/2)$.

MLP for multi-class classification: KL Divergence

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- Does KL-divergence obey triangle inequality?

MLP for multi-class classification: KL Divergence

Some useful properties of KL-divergence:

- For two discrete probability distributions $p = (p_1, p_2, \dots, p_C)$ and $q = (q_1, q_2, \dots, q_C)$, $q_j > 0$, $\forall j = 1, \dots, C$, $KL(p||q) \geq 0$.

MLP for multi-class classification: KL Divergence

KL-Divergence: Equivalent Representation

- Given two discrete probability distributions $p = (p_1, \dots, p_C)$ and $q = (q_1, \dots, q_C)$, where $q_j > 0 \forall j = 1, \dots, C$, KL-divergence between p and q is defined as:

$$KL(p||q) = \sum_{j=1}^C p_j \log \frac{p_j}{q_j} = \sum_{j=1}^C p_j \log p_j - \sum_{j=1}^C p_j \log q_j.$$

MLP for multi-class classification: KL Divergence

KL-Divergence: Equivalent Representation

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Note: $\sum_{j=1}^C p_j \log p_j$ is called **negative entropy** associated with distribution p (denoted by $NE(p)$) and $-\sum_{j=1}^C p_j \log q_j$ is called **cross-entropy** between p and q (denoted by $CE(p, q)$).

- Hence $KL(p||q) = NE(p) + CE(p, q)$.

MLP for multi-class classification

Question:

- Why should we transform $y \in \mathcal{Y}$ taking an integer value, to a $y^{onehotenc}$ which represents a discrete probability distribution?

One possible answer:

- If the prediction $\hat{y}$ made by MLP is also a discrete probability distribution, then the comparison between $\hat{y}$ and $y^{onehotenc}$ becomes a comparison between two discrete probability distributions.
- Multiple ways to compare two probability distributions.
- Loss function design becomes possibly simpler?

MLP for multi-class classification

Loss function using KL-Divergence:

- Given two discrete probability distributions $p = (p_1, \dots, p_C)$ and $q = (q_1, \dots, q_C)$, where $q_j > 0 \forall j = 1, \dots, C$, KL-divergence between p and q is defined as:

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- Our aim is to minimize the error measured using KL-divergence between the true distribution p and the predicted distribution q .

MLP for multi-class classification

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- However, minimizing KL-divergence between p and q is equivalent to minimizing cross-entropy between p and q . (why?)

MLP for multi-class classification

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- Our aim is to minimize the error measured using KL-divergence between the true distribution p and the predicted distribution q .
- However, minimizing KL-divergence between p and q is equivalent to minimizing cross-entropy between p and q . (why?)
- Hence we will consider the cross-entropy between p and q as our loss function.

MLP for multi-class classification

Cross-entropy loss function:

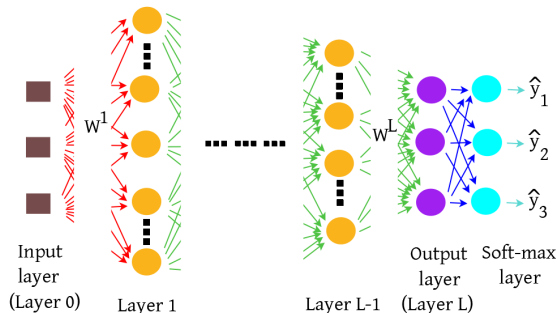
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$$CE(p, q) = - \sum_{j=1}^C p_j \log q_j.$$

MLP for multi-class classification

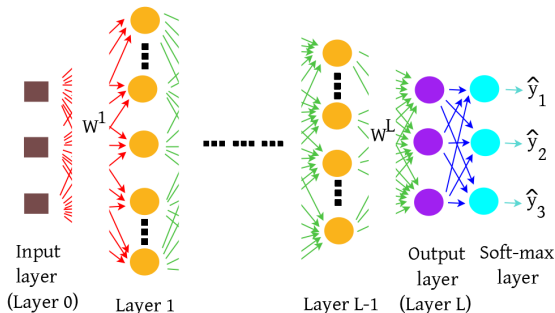
- **Input:** Training Data $D = \{(x^i, y^i)\}_{i=1}^S$, $x^i \in \mathcal{X} \subseteq \mathbb{R}^d$, $y^i \in \tilde{\mathcal{Y}}$, $\forall i \in \{1, \dots, S\}$ and MLP architecture parametrized by weights w .
- Without loss of generality, assume $\tilde{\mathcal{Y}} = \{0, 1\}^C$ corresponding to the output space $\mathcal{Y} = \{1, \dots, C\}$, $C \geq 2$ and y^i are one-hot encoded vectors.

MLP for multi-class classification



- Input:** Training Data $D = \{(x^i, y^i)\}_{i=1}^S$, $x^i \in \mathcal{X} \subseteq \mathbb{R}^d$, $y^i \in \tilde{\mathcal{Y}}$, $\forall i \in \{1, \dots, S\}$ and MLP architecture parametrized by weights w .
- Aim of training MLP:** To learn a parametrized map $h_w : \mathcal{X} \rightarrow \tilde{\mathcal{Y}}$ such that for the training data D , we have $y^i = h_w(x^i)$, $\forall i \in \{1, \dots, S\}$.
- Aim of using the trained MLP model:** For an unseen sample $\hat{x} \in \mathcal{X}$, predict $\hat{y} = h_w(\hat{x}) = \text{MLP}(\hat{x}; w)$; to find integer label use $\text{argmax}_j \hat{y}_j$.

MLP for multi-class classification



Methodology for training MLP

- Design a suitable loss (or error) function $e: \tilde{\mathcal{Y}} \times \tilde{\mathcal{Y}} \rightarrow [0, +\infty)$ to compare the actual label y^i and the prediction $\hat{y}^i$ made by MLP using $e(y^i, \hat{y}^i), \forall i \in \{1, \dots, S\}$.
- recall:** $e(y^i, \hat{y}^i) = \text{CE}(y^i, \hat{y}^i)$, the cross-entropy loss function.
- Usually the error is parametrized by the weights w of the MLP and is denoted by $e(\hat{y}^i, y^i; w)$.
- Use Gradient descent/SGD/mini-batch SGD to minimize the total error:

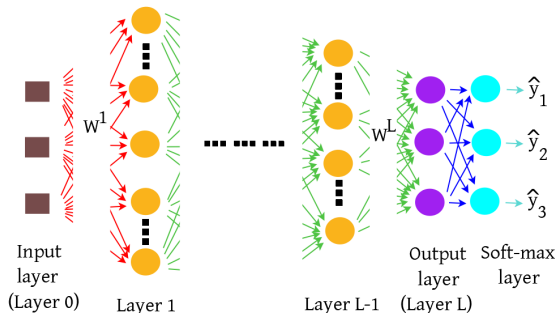
$$E = \sum_{i=1}^S e(\hat{y}^i, y^i; w) =: \sum_{i=1}^S e^i(w).$$

SGD for training MLP for multi-class classification

SGD Algorithm to train MLP

- **Input:** Training Data $D = \{(x^i, y^i)\}_{i=1}^S$, $x^i \in \mathcal{X} \subseteq \mathbb{R}^d$, $y^i \in \tilde{\mathcal{Y}}$, $\forall i$;
MLP architecture, max epochs K , learning rates γ_k , $\forall k \in \{1, \dots, K\}$.
- Start with $w^0 \in \mathbb{R}^d$.
- For $k = 0, 1, 2, \dots, K$
 - ▶ Choose a sample $j_k \in \{1, \dots, S\}$.
 - ▶ Find $\hat{y}^{j_k} = \text{MLP}(x^{j_k}; w^k)$. (forward pass)
 - ▶ Compute error $e^{j_k}(w^k)$ using cross-entropy loss function.
 - ▶ Compute error gradient $\nabla_w e^{j_k}(w^k)$ using backpropagation.
 - ▶ Update: $w^{k+1} \leftarrow w^k - \gamma_k \nabla_w e^{j_k}(w^k)$.
- **Output:** $w^* = w^{K+1}$.

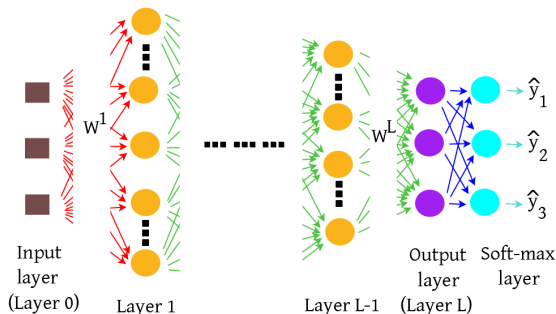
Training MLP for multi-class classification



Recall forward pass: For an arbitrary sample (x, y) from training data D , and the MLP with weights $w = (W^1, W^2 \dots, W^L)$, the prediction $\hat{y}$ is computed using forward pass as:

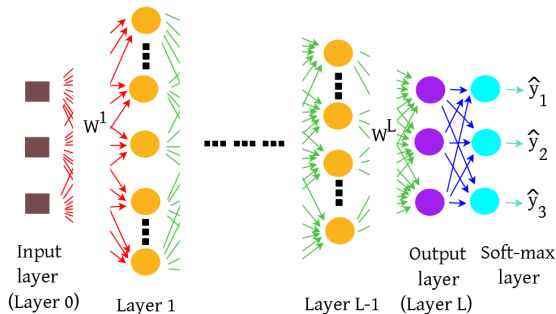
$$\hat{y} = \text{MLP}(x; w) = \text{softmax}(\phi(W^L \phi(W^{L-1} \dots \phi(W^1 x) \dots))).$$

Training MLP for multi-class classification



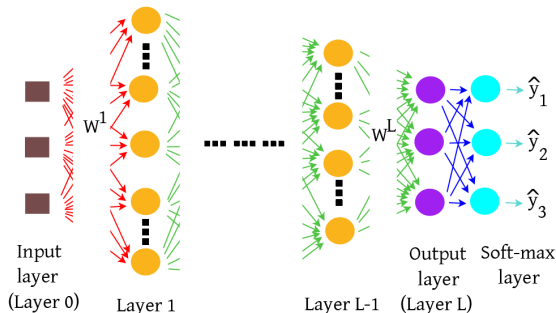
Backpropagation: For an arbitrary sample (x, y) from training data D , and the MLP with weights $w = (W^1, W^2 \dots, W^L)$, how do we compute the error gradient?

Training MLP for multi-class classification



Backpropagation: Indeed, the error gradients are very similar to the binary classification setup.

Training MLP for multi-class classification



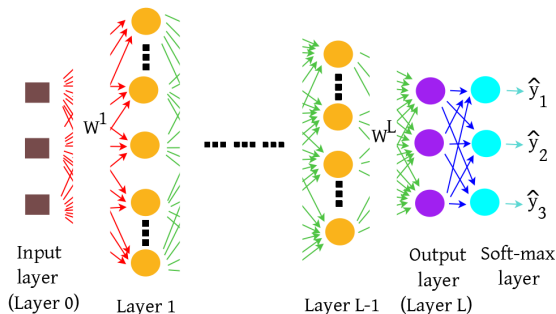
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For an arbitrary sample (x, y) from training data D , and the MLP with weights $w = (W^1, W^2, \dots, W^L)$, the error gradient with respect to weights W^ℓ in ℓ -th layer is:

$$\nabla_{W^\ell} e = \text{Diag}(\phi^{\ell'}) \delta^\ell (a^{\ell-1})^\top$$

$$\text{where } \text{Diag}(\phi^{\ell'}) = \begin{bmatrix} \phi'(z_1^\ell) & & \\ & \ddots & \\ & & \phi'(z_{N_\ell}^\ell) \end{bmatrix}, \quad \delta^\ell = \begin{bmatrix} \frac{\partial e}{\partial a_1^\ell} \\ \vdots \\ \frac{\partial e}{\partial a_{N_\ell}^\ell} \end{bmatrix} \quad \text{and} \quad a^{\ell-1} = \begin{bmatrix} a_1^{\ell-1} \\ \vdots \\ a_{N_{\ell-1}}^{\ell-1} \end{bmatrix}.$$

Training MLP for multi-class classification



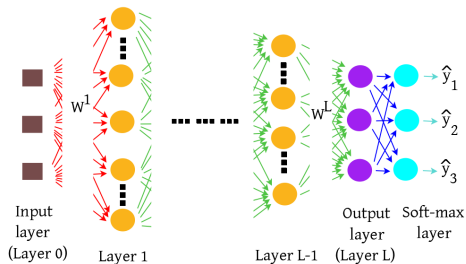
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$$\text{where } V^{\ell+1} = (W^{\ell+1})^\top \text{Diag}(\phi^{\ell+1'}).$$

Training MLP for multi-class classification



Backpropagation: Indeed, the error gradients are very similar to the binary classification setup.

For an arbitrary sample (x, y) from training data D , and the MLP with weights $w = (W^1, W^2, \dots, W^L)$, the error gradient with respect to weights W^ℓ in ℓ -th layer is:

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The main difference arises in the procedure to find $\delta^L = \begin{bmatrix} \frac{\partial e}{\partial a_1^L} \\ \vdots \\ \frac{\partial e}{\partial a_C^L} \end{bmatrix}$.

Training MLP for multi-class classification: Finding error gradients at the output layer

Recall: For an arbitrary sample (x, y) from training data D , and the prediction $\hat{y}$ from the MLP, we know the following:

- y is in one-hot encoded form: $y = \begin{bmatrix} y_1 \\ \vdots \\ y_C \end{bmatrix}$ with some particular y_c taking value 1 (corresponding to the label c) and $y_j = 0, j \neq c$.

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- y and $\hat{y}$ are discrete probability distributions.

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- $\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_C \end{bmatrix}$.

- $e(\hat{y}, y)$ is the cross-entropy error function: $e(\hat{y}, y) = \sum_{j=1}^C e^j = -\sum_{j=1}^C y_j \log \hat{y}_j$.

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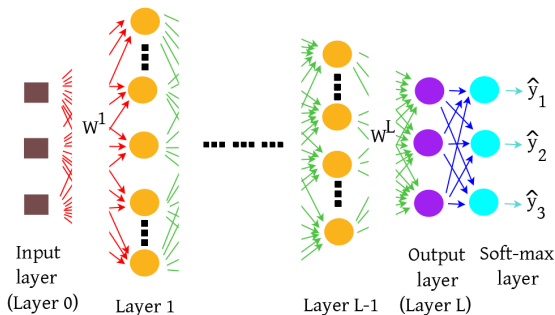
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- **Note:** $\hat{y}_j = \frac{\exp(a_j^L)}{\sum_{r=1}^C \exp(a_r^L)}, \forall j = 1, \dots, C$.

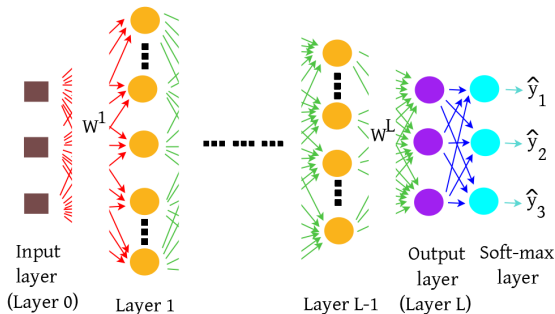
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- $e(\hat{y}, y)$ is the cross-entropy error function: $e(\hat{y}, y) = \sum_{j=1}^C e^j = - \sum_{j=1}^C y_j \log \hat{y}_j$, with

$$\hat{y}_j = \frac{\exp(a_j^L)}{\sum_{r=1}^C \exp(a_r^L)}, \forall j = 1, \dots, C.$$

Training MLP for multi-class classification: Finding error gradients at the output layer

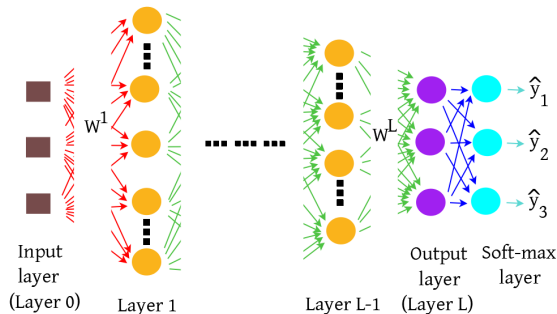


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Aim: To find $\delta^L = \left[\frac{\partial e}{\partial a_1^L} \quad \dots \quad \frac{\partial e}{\partial a_C^L} \right]^\top$.

Training MLP for multi-class classification: Finding error gradients at the output layer



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Training MLP for multi-class classification: Finding error gradients at the output layer

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- $\frac{\partial e^m}{\partial \hat{y}_m} = -\frac{y_m}{\hat{y}_m}$.
- $\frac{\partial \hat{y}_m}{\partial a_j^L} = \frac{\partial}{\partial a_j^L} \left(\frac{\exp(a_m^L)}{\sum_{r=1}^C \exp(a_r^L)} \right)$.

Training MLP for multi-class classification: Finding error gradients at the output layer

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Training MLP for multi-class classification: Finding error gradients at the output layer

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- We have $\frac{\partial e}{\partial a_j^L} = \sum_{m=1}^C \frac{\partial e^m}{\partial \hat{y}_m} \frac{\partial \hat{y}_m}{\partial a_j^L}$.
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- Hence by substitution, $\frac{\partial e}{\partial a_j^L} = \sum_{m=1}^C \sum_{m \neq j} y_m \hat{y}_j - y_j(1 - \hat{y}_j) = \hat{y}_j - y_j$.